

Geography Ch-11 (Must-Do) — Islands & Coral Reefs

Must-Do / Foundation+Standard | GS-I (Physical Geography) | Sources: GC Leong + Majid Husain
Island types; coral reef types & Darwin's subsidence theory (fringing -> barrier -> atoll)
CA anchor: 4th global coral bleaching event (2024)

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

FOUNDATION — Island Types & Coral Reef Formation (GC Leong / Majid)

GS-I • Physical Geography → Islands & Coral Reefs (Source: GC Leong, Majid Husain)

■ NEWS TRIGGER

Coral reefs — India's Lakshadweep, Gulf of Mannar and Andaman & Nicobar — are biodiversity hotspots now hit by global bleaching, making reef geomorphology exam-hot.

■ INTRO & ORIGIN

Islands are classed as **continental** (detached from a landmass), **oceanic** (volcanic, rising from the deep) or **coral**. Reefs are built by tiny **coral polyps** under strict conditions.

*Corals need **warm (23-25 C)**, **shallow**, **clear**, **salty**, **sediment-free** water. Reefs grow as **fringing**, **barrier** or **atoll** — explained best by Darwin's subsidence theory.*

■ KEY DATA

Category	Type	Example
Islands	Continental	British Isles, Newfoundland
Islands	Oceanic (volcanic)	Hawaii; Barren Island (India)
Islands	Coral	Maldives, Laccadives (Lakshadweep); Marshall Is.
Reefs	Fringing reef	Attached to shore, no/narrow lagoon
Reefs	Barrier reef	Offshore, wide lagoon (Great Barrier Reef)
Reefs	Atoll	Ring reef enclosing a lagoon (Maldives)

■ STATIC THEORY

- **Coral polyps** are tiny marine animals whose limestone skeletons build reefs; they host symbiotic algae (zooxanthellae).
- **Ideal conditions:** warm tropical water (~23-25 C), shallow (**within ~sunlit depth**), clear and sediment-free, and normal salinity — hence reefs cluster between ~30 N-30 S.
- **Three reef types:** **fringing** (against the shore), **barrier** (separated by a wide lagoon), and **atoll** (circular reef around a lagoon).
- **Darwin's subsidence theory:** reefs begin as fringing reefs around a volcanic island; as the island slowly **subsides**, upward coral growth turns it into a barrier reef and finally an **atoll**.
- **Daly's theory** stresses sea-level (glacial) change; borings show basalt at depth (supporting Darwin) — so a combination explains reefs best (GC Leong).

■ MUST-KNOW FACTS

1. Island types: continental, oceanic (volcanic), coral.
2. Coral needs warm (~23-25 C), shallow, clear, sediment-free salt water.
3. Reef types: fringing -> barrier -> atoll.

■ UPSC TRAPS

- ✗ Corals thrive in cold, deep, muddy water.
- ✓ Corals need WARM, SHALLOW, CLEAR, sediment-free saline water.
- ✗ An atoll forms before a barrier reef.

4. Darwin's SUBSIDENCE theory: fringing -> barrier -> atoll as island sinks. 5. Zooxanthellae = symbiotic algae; their loss = bleaching. 6. Atoll = ring reef enclosing a lagoon (Maldives).	✓ Order (Darwin): fringing -> barrier -> ATOLL (as the island subsides). ✗ Coral islands are high and rugged. ✓ Coral islands are LOW, emerging just above the water; volcanic islands are high.
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■ MAINS ANGLE Explain the conditions for coral growth and Darwin's subsidence theory, applying reef types to India's Lakshadweep atolls and Gulf of Mannar fringing reefs.	■ STUDY LINK Geography → Physical → Islands & Coral Reefs (GC Leong: Coral Reefs; Majid: Islands)
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■ MEDIUM UPSC RELEVANCE

MEDIUM CA ANCHOR — Fourth Global Coral Bleaching Event (2024)	GS-I / GS-III · Coral biology applied → Climate stress
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■ NEWS TRIGGER In 2024 NOAA declared the FOURTH global coral bleaching event — warm seas (global warming + El Nino) bleached reefs worldwide, including India's Lakshadweep and Gulf of Mannar.

■ INTRO & ORIGIN Bleaching is heat stress made visible: corals expel their zooxanthellae, turn white and, if stress persists, die — directly linking reef geomorphology to climate change.
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■ STATIC THEORY • Bleaching = expulsion of symbiotic zooxanthellae under heat stress; the coral whitens and may die. • Trigger 2024: record sea-surface temperatures (warming + El Nino); the 4th global event (after 1998, 2010, 2015-17). • Indian hotspots hit: Lakshadweep (severe), Gulf of Mannar (India's first marine biosphere reserve, ~117 coral species).

■ MUST-KNOW FACTS 1. 2024 = 4th global bleaching event (NOAA); after 1998, 2010, 2015-17. 2. Cause: high SST (warming + El Nino) -> zooxanthellae expelled. 3. India hit: Lakshadweep, Gulf of Mannar, Andaman & Nicobar.	■ UPSC TRAPS ✗ Bleached corals are already dead. ✓ Bleaching is stress (algae expelled); corals can RECOVER if temperatures fall soon enough.
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■ MAINS ANGLE Link coral biology to ocean warming and the case for marine protected areas / emission cuts.	■ STUDY LINK Geography → Coral Reefs → Bleaching / Climate (applied CA)
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